

Amir Khajavirad

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Summary

Biomedical Engineer working at the intersection of algorithms, robotics, and experimental systems. I build data driven systems that operate under real world conditions where signals are noisy, components are imperfect, and reliability is critical. My work focuses on processing multi modal data, designing algorithmic pipelines, and integrating hardware and software into systems that produce consistent and interpretable results.

Technical Skills

Programming and Data: Python, SQL, Bash, Linux

Algorithms and Data Analysis: Algorithm Design, Multivariate Data Analysis, Statistical Modeling, Experimental Design

Signal Processing: Filtering, Noise Reduction, Feature Extraction, Sensor Signal Interpretation

Machine Learning: Scikit learn, XGBoost, SHAP, Feature Engineering, Model Optimization

Applied AI: LLM Integration, Prompt Design, Workflow Automation

Systems and Robotics: Embedded Systems, Raspberry Pi, Sensor Integration, Control Systems, Data Acquisition

Engineering and Design: SolidWorks, CAD Modeling, Mechanical Design, Prototyping

Software: Object Oriented Programming, Git

Mathematics: Linear Algebra, Probability, Statistics

Experience

Robotics Engineer Intern

Nov 2025 – Present

University of Alberta — Edmonton, AB

- Engineered an automated robotic platform integrating embedded hardware, sensors, and real time control systems for behavioral experiments
- Designed and implemented classification pipelines for multi modal sensor data including RFID, photo interrupt sensors, and load cells
- Processed large multivariate datasets using filtering, noise reduction, and feature extraction to improve signal reliability
- Improved system stability and reduced failure cases by 30 percent through optimization of control logic and sensor feedback loops
- Reduced system errors by 25 percent through debugging, calibration, and refinement of hardware and software interactions
- Automated experimental workflows, reducing manual intervention by 60 to 70 percent and enabling repeatable data collection
- Developed TCP based communication protocol for synchronized multi robot operation and centralized control
- Designed system architecture for reliable integration across multiple hardware and software components
- Integrated RFID, photo interrupt sensors, and load cells for real time event detection and outcome classification
- Achieved real time system response with latency below 100 milliseconds
- Built and validated electromechanical prototypes for reliable experimental data acquisition

Projects

Standardized Biomechanical Testing Platform for Distal Biceps Fixation

- Designed and built a biomechanical testing platform for controlled mechanical data collection and performance analysis
- Developed 3D printed anatomical models and custom fixtures integrated with a BOSE ElectroForce system
- Implemented cyclic loading and pull to failure protocols based on biomechanical literature
- Engineered reproducible testing workflows to minimize experimental variability
- Enabled quantitative comparison of fixation methods using controlled data acquisition and analysis

PredictIQ Football Match Outcome Prediction (XGBoost)

github.com/shervinafzali/PredictIQ

- Developed an end to end machine learning pipeline including data extraction, feature engineering, and model training
- Designed time aware validation strategies to simulate real world prediction conditions

- Optimized an XGBoost model achieving a Macro F1 score of 0.357 and outperforming baseline models
- Applied SHAP for feature importance analysis and model interpretability

Breast Cancer Subtype Classification

- Built classification models on high dimensional gene expression data with over 54000 features
- Applied dimensionality reduction techniques including PCA, Kernel PCA, and LDA
- Evaluated multiple models including SVM, Random Forest, KNN, and Logistic Regression
- Improved model robustness through outlier detection and preprocessing

Adaptive Circadian Lighting System for Neonatal Incubators

- Designed and implemented an embedded biomedical system for circadian aligned lighting control
- Developed microcontroller based control logic and integrated electronic components
- Built and validated a complete hardware prototype through testing and debugging

Education

Master of Science, Biomedical Engineering

Jan 2024 – Dec 2025

University of Alberta — Edmonton, AB

- Developed a biomechanical testing platform integrating CAD design, 3D printed models, and mechanical testing systems

Bachelor of Science, Biomedical Engineering

Sep 2017 – Jun 2022

Islamic Azad University — Tehran, Iran

- Designed and built an embedded biomedical device for neonatal circadian lighting control

Additional Experience

Recreation Facility Operator

Sep 2024 – Present

University of Alberta — Edmonton, AB

- Managed operations in a high traffic environment, ensuring safety and real time issue resolution
- Maintained operational reliability under dynamic conditions

Leadership and Activities

- DRA Research Supervisor, University of Alberta Supervised undergraduate research activities and supported development of experimental systems
- Event Coordination Volunteer, Engineering Graduate Research Symposium, University of Alberta
- Event Volunteer, BioAlberta Health and Life Sciences Showcase 2025